

Algebra 1
Unit 2
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

1. Rewrite the expression in each step using the description: $6(x + 4) + 3x$

$6x + 24$	Distribute the 6 to $(x + 4)$.
$6x + 24 + 3x$ $9x + 24$	Combine like terms.

2. Rewrite the expression in each step using the description: $-3x + 5(-2x - 5)$

$-10x - 25$	Distribute the 5 to $(-2x - 5)$.
$-3x - 10x - 25$ $-13x - 25$	Combine like terms.

Use the order of operations to simplify each polynomial.

3. $3(x + 5) + 2(x - 8)$
 $3x + 15 + 2x - 16$
 $5x - 1$

4. $\frac{3}{5}(x - 6) + 3x$

$$\frac{3}{5}x - \frac{18}{5} + 3x$$

$$\frac{18}{5}x - \frac{18}{5}$$

5. $-2(x^2 + 4x) + 7(x - 2)$

$$-2x^2 - 8x + 7x - 14$$

$$-2x^2 - x - 14$$

6. $2x^2 - 45(x - 7)$

$$2x^2 - 45x + 315$$

7. $(x + 3)(x - 5) - 9(2x^2 + 3x)$

$$x^2 - 5x + 3x - 15 - 18x^2 - 27x$$

$$-17x^2 - 29x - 15$$

8. $-\frac{1}{3}(6x - 12) + 7x$

$$-2x + 4 + 7x$$

$$5x + 4$$

9. $2x^2 + (x + 5)(x - 5)$

$$2x^2 + x^2 - 5x + 5x - 25$$

$$3x^2 - 25$$

10. $3x(x - 4) + 7(x - 3)$

$$3x^2 - 12x + 7x - 21$$

$$3x^2 + -5x - 21$$